NIGERIA DHS–H3 INTEGRATION PROJECT

Federal Capital Territory (FCT) Pilot

Environmental Exposure, Built-Environment Morphology and Demographic & Health Survey Integration

Document type	Client technical briefing & deliverables guide
Geographic scope	Federal Capital Territory (FCT), Nigeria
Analytical units	H3 hexagonal tessellation + 37 DHS clusters
Reference year	2024 environmental layers; DHS 2023–24 round
Platforms used	Google Earth Engine · Google Colab · Python (GeoPandas, Folium)
Date prepared	12 October 2026

This report explains the complete FCT pilot workflow, the data that were integrated, the spatial framework used, the interactive map and Excel deliverables, and how the client should interpret and use the outputs.

1. Introduction and Purpose

The Federal Capital Territory (FCT) pilot is the first operational demonstration of a national framework that links three families of information at a consistent spatial scale:

- **Environmental exposure** — rainfall, land-surface temperature (LST), vegetation greenness (NDVI) and elevation.
- **Built-environment morphology** — building counts, building footprint area, and road length within hexagonal cells.
- **Demographic and Health Survey (DHS) indicators** — household size, wealth, urban/rural residence, electricity access, and sample counts from the household (HR), women's (IR), children's (KR), men's (MR) and household-member (PR) recodes.

The pilot answers a practical question: can environmental and built-environment measures be extracted and attached to official DHS clusters in a transparent, reproducible way that is ready for analysis and client presentation? The answer demonstrated here is yes. The same workflow is designed to scale from FCT to other Nigerian states and ultimately to a national product.

All processing was performed with open, documented tools. Environmental rasters were prepared in **Google Earth Engine (GEE)**. Spatial joins, zonal statistics, survey aggregation and the interactive map were produced in **Google Colab** using the Python scientific stack (GeoPandas, rasterio/rasterstats, Folium, Pandas). Outputs are delivered as an interactive HTML map, an Excel table, a GeoPackage and supporting CSV files.

2. Spatial Framework: H3 Tessellation and DHS Clusters

2.1 H3 hexagonal tessellation

The analytical backbone is the **Uber H3** discrete global grid. H3 partitions the Earth's surface into hexagons of approximately equal area. Two resolutions were used in the pilot:

- **H3 resolution 8** — coarser cells, suitable for regional environmental context and for linking to DHS clusters.
- **H3 resolution 9** — finer cells, used where building and road morphology required higher detail.

The FCT H3 master layer contains **5,634 hexagonal cells** covering the Territory. Each cell carries a unique **H3_ID** and is the unit at which environmental and morphology statistics were calculated. Using a regular tessellation avoids the arbitrary shapes of administrative boundaries and makes results comparable across cities and states.

2.2 DHS clusters

The Nigeria DHS provides geocoded cluster locations (displaced for confidentiality according to DHS rules). For the FCT pilot, **37 DHS clusters** fall within or are assigned to the Territory. Each cluster has:

- A permanent identifier (**DHSCLUST** / **DHSID**).
- Latitude and longitude (displaced coordinates).
- Pre-assigned **H3_8_ID** and **H3_9_ID** linking it to the tessellation.
- Urban/rural classification and administrative labels.

Because DHS coordinates are intentionally displaced, environmental and morphology values attached to a cluster represent the **context of the H3 cell** that contains the published point, not the precise location of every household. This is the standard, accepted interpretation in DHS–remote-sensing research.

3. Data Components, Sources and Applications

3.1 Environmental variables

Variable	Source dataset	Platform / processing	Unit & meaning
Annual rainfall	CHIRPS Daily (UCSB-CHG)	Google Earth Engine → annual sum → GeoTIFF; point sampling at clusters in Colab	Millimetres (mm). Total precipitation for calendar year 2024.
Land surface temperature (LST)	MODIS MOD11A2 (Collection 6.1)	GEE → annual mean of daytime LST → scale to °C → GeoTIFF; point sampling in Colab	Degrees Celsius (°C). Annual mean daytime surface temperature.
NDVI	Landsat 8/9 Collection 2 Level-2	GEE → cloud filter <20% → scale reflectance → mean NDVI → GeoTIFF	Unitless (–1 to +1). Annual mean vegetation greenness.

Variable	Source dataset	Platform / processing	Unit & meaning
Elevation	Copernicus DEM GLO-30	GEE → mosaic → clip to FCT → GeoTIFF	Metres above sea level. Mean elevation within each H3 cell / at cluster point.

Table 1. Environmental variables, sources and processing route.

Environmental rasters were first exported from GEE as GeoTIFFs. Because CHIRPS and MODIS are relatively coarse, values for the 37 clusters were obtained by **direct point sampling** at cluster coordinates (most reliable for coarse rasters) with zonal statistics as a secondary check. NDVI and elevation, being finer, had near-complete coverage from H3 zonal statistics.

3.2 Built-environment morphology

Building footprints and road networks were summarised inside every H3 cell. The resulting metrics include:

- **Building_Count** / H3_8_Building_Count — number of building footprints.
- **Total_Building_Area_km2** — total footprint area in square kilometres.
- **Total_Road_Length_km** / H3_8_Total_Road_Length_km — total length of road centre-lines.
- **Road_Segment_Count** — number of road segments.

These morphology statistics were produced in earlier project stages (building and road extraction, H3 intersection) and stored in the FCT Integrated Morphology GeoPackage. They describe the intensity of the built environment surrounding each DHS cluster.

3.3 DHS survey modules

Module	File / recode	What it contributes at cluster level
HR — Household	DHS_Household_Level_Integrated / HR_Selected	Household count, mean household size, wealth score & quintile, urban/rural share, electricity access, modal water/toilet/housing materials.
IR — Women	DHS_IR_Woman_Level	Number of women interviewed in the cluster; mean age and wealth summaries where available.
KR — Children	DHS_KR_Child_Level	Number of children (birth records) linked to the cluster.
MR — Men	DHS_MR_Man_Level	Number of men interviewed in the cluster.
PR — Members	DHS_PR_Household_Member_Level	Total household members listed in the cluster.

Table 2. DHS modules used and the cluster-level information derived from each.

Household-level records were aggregated to the 37 FCT clusters using the standard DHS cluster identifier (hv001 / DHSCLUST). Only clusters that intersect the FCT spatial framework receive environmental and morphology values; the national household file remains intact for broader analysis.

4. Methodological Framework and Steps

The pilot followed a fixed sequence of stages. Each stage has a clear input, process and output so that the work can be audited and repeated for other states.

Step	Stage name	Platform	Main output
1	Boundary & H3 tessellation	Colab / H3 library	FCT H3-8 and H3-9 polygons with unique IDs
2	Building & road morphology	Colab + spatial intersection	Building counts, area, road length per H3 cell
3	DHS GPS & cluster–H3 assignment	Colab	37 FCT clusters with H3_8_ID / H3_9_ID
4	Environmental raster preparation	Google Earth Engine	GeoTIFFs: rainfall, LST, NDVI, elevation
5	Zonal & point extraction	Colab (rasterstats / rasterio)	Environmental values on H3 cells and at cluster points
6	DHS household linkage (HR–IR–KR–MR–PR)	Colab	One-record-per-household integrated file + QC

Step	Stage name	Platform	Main output
7	Cluster-level socioeconomic aggregation	Colab	Wealth, household size, sample counts per cluster
8	Master integration & map	Colab (Folium)	Interactive HTML map + Excel + GeoPackage

Table 3. End-to-end pilot workflow (framework).

Design principles applied throughout: (a) every join is performed on documented identifiers (H3_ID or DHSCLUST), never on row order; (b) environmental values are never assigned from a whole-FCT average to individual clusters; (c) DHS confidentiality displacement is respected—results describe cluster context, not exact household exposure; (d) all scripts and intermediate files are retained so the pilot is fully reproducible.

5. Results Summary for the FCT Pilot

After integration, the pilot produces a coherent analytical table at the **cluster level** (37 rows) and a supporting H3-level environmental–morphology layer (5,634 cells).

Indicator	Value / coverage
H3 cells with environmental attributes	5,634
DHS clusters in FCT pilot	37
Clusters with rainfall (point sampling)	37 / 37
Clusters with LST	37 / 37
Clusters with NDVI	35 / 37
Clusters with elevation	35 / 37
Clusters with building & road morphology	Linked via H3_8_ID
Household records matched to FCT clusters	Approx. 1,000+ (FCT subset of national HR)
Socioeconomic fields at cluster level	Household size, wealth score/quintile, urban/rural, electricity, sample counts (HR/IR/KR/MR/PR)

Table 4. Key quantitative outcomes of the FCT pilot.

Rainfall and LST, which come from coarser sensors, achieved complete coverage at the 37 cluster points through direct point sampling. NDVI and elevation, derived from finer rasters, are available for 35 of 37 clusters. Morphology metrics are inherited from the H3 cell that contains each cluster.

6. How to Read and Use the Interactive HTML Map

The primary visual deliverable is an interactive map saved as a single HTML file. It opens in any modern web browser (Chrome, Firefox, Edge) with no software installation and no internet connection required after download (basemaps need a connection on first load).

6.1 File location

Path: NIGERIA_DHS_H3_PROJECT / 08_Environmental_Integration / Maps / FCT_Pilot_Full_Integration_Map_with_Roads.html

6.2 Map layers

- **Basemaps** — OpenStreetMap, Esri Satellite, Esri Topographic (switch in the layer control).
- **H3 thematic layers** — NDVI, Rainfall, LST, Elevation, Building Count, Building Area, Road Length. Toggle each on/off; only one or two need be visible at a time for clarity.
- **DHS Clusters (37)** — red circular markers. Click any marker to open a detailed pop-up.
- **H3 Cell Boundaries** — optional outline layer for orientation.

6.3 Cluster pop-up content

Each cluster pop-up is organised in four blocks:

• **Socioeconomic & household profile** — household record count, mean household size, wealth score and quintile, urban/rural share, electricity access.
• **Environmental characteristics** — rainfall (mm), LST (°C), NDVI, elevation (m).
• **Built-environment morphology** — building count, building area, road length (and road segments where available).
• **Survey sample sizes** — counts of households (HR), women (IR), children (KR), men (MR) and members (PR) linked to that cluster.

6.4 Practical tips for the client

• Use the layer control (top-left) to turn layers on and off; start with NDVI or Rainfall plus the DHS clusters.
• Click a red cluster point rather than hovering, to keep the full pop-up open while you read.
• The legend is fixed at the bottom-right so it does not cover the title or the layer list.
• Fullscreen mode (icon top-left) is useful for presentations.
• The map can be projected live in a meeting or sent as a self-contained file to partners.

7. How to Use the Excel Deliverable

All numeric attributes attached to the 37 clusters are also delivered as a single Excel workbook. This is the table that analysts will use for statistics, dashboards or further modelling.

7.1 File location

Path: NIGERIA_DHS_H3_PROJECT / 08_Environmental_Integration / FCT_DHS_Clusters_Full_Integrated.xlsx

7.2 Structure of the workbook

• One sheet, one row per DHS cluster (37 rows).
• Columns are ordered so that identifiers and the most important environmental, morphology and socioeconomic fields appear first.
• No geometry is stored in Excel; coordinates and H3 IDs are retained as attributes for later GIS use.

7.3 Recommended use

• Filter or sort by urban/rural, wealth quintile or rainfall band.
• Correlate NDVI or LST with wealth or household size.
• Export selected columns into statistical packages (Stata, R, SPSS) or BI tools (Power BI, Tableau).
• Join back to the GeoPackage if maps need to be redrawn in QGIS or ArcGIS.

A companion CSV and GeoPackage with the same attribute table are also provided for users who prefer those formats.

8. Deliverables Package

Deliverable	Format	Location / name	Purpose
Interactive map	HTML	Maps/FCT_Pilot_Full_Integration_Map_with_Roads.html	Exploration, presentation, client walkthrough
Cluster attribute table	Excel (.xlsx)	FCT_DHS_Clusters_Full_Integrated.xlsx	Analysis, reporting, handover to statisticians
Cluster attribute table	CSV	FCT_DHS_Clusters_Full_Integrated.csv	Lightweight exchange, scripts
Spatial cluster layer	GeoPackage	FCT_DHS_Clusters_Full_Integrated.gpkg	GIS mapping and further spatial analysis
H3 environmental master	GeoPackage + CSV	FCT_H3_Environmental_Master.*	Full tessellation with environment & morphology
Cluster environmental (improved)	CSV + GPKG	FCT_DHS_Clusters_with_Environmental_Improved.*	Traceable intermediate product
This client report	PDF	FCT_Pilot_Client_Report.pdf	Documentation and decision support

Table 5. Complete list of pilot deliverables.

9. Interpretation Notes and Limitations

- **DHS displacement.** Published cluster coordinates are displaced (typically up to 2 km urban, 5 km rural). Environmental and morphology values therefore describe the H3 neighbourhood of the published point, not the exact dwelling.
- **Coarse climate rasters.** CHIRPS (~5.5 km) and MODIS LST (~1 km) cannot resolve micro-climates inside a single neighbourhood. They are appropriate for between-cluster and between-city comparison.
- **Temporal alignment.** Environmental layers represent calendar year 2024. DHS fieldwork spans the 2023–24 survey round. Exact day-of-interview climate is not claimed.
- **FCT-only environmental coverage.** The present rasters and morphology are for FCT. National extension requires the same GEE and morphology pipeline for additional states.
- **Category codes.** Water source, toilet and housing-material fields are DHS codes. They must be interpreted with the official DHS recode manual before being labelled “improved” or “unimproved”.

10. Reproducibility and Platforms

Every major step is scripted. The environmental GeoTIFFs were generated with version-controlled GEE JavaScript; all subsequent joins, zonal statistics, survey aggregation and map generation are Python notebooks run in Google Colab. Intermediate products (H3 morphology, cluster–H3 crosswalk, household integrated file, QC reports) are stored under the project folder structure:

- 01–05 — Boundary, H3, morphology and integrated morphology
- 06 — DHS GPS, cluster–H3 assignment and master cluster table
- 07 — DHS survey extraction, linkage and QC
- 08 — Environmental integration, maps and final tables

A third party with access to the same Drive folder and the published scripts can regenerate the HTML map and Excel table without undocumented manual steps.

11. Recommended Next Steps

- Validate a small sample of cluster pop-ups against the Excel rows in a joint review meeting.
- Extend the identical GEE + Colab pipeline to one additional state (e.g. Lagos or Kano) to prove national scalability.
- Add survey weights to cluster-level socioeconomic summaries where formal inference is required.
- Optional: publish a short methods note (1–2 pages) alongside the map for academic or donor audiences.
- Archive the final HTML, Excel and GeoPackage with a version tag and date for contractual handover.

12. Closing Statement

The FCT pilot demonstrates that environmental exposure, built-environment morphology and DHS socioeconomic indicators can be integrated at the cluster and H3-cell scales in a transparent, reproducible workflow. The interactive map allows non-technical stakeholders to explore spatial patterns; the Excel table gives analysts a ready-to-use dataset; and the documented pipeline provides a clear path from FCT to a national product.

All primary deliverables are already in the project Drive folder. This report serves as the accompanying explanation of what was done, why it was done that way, and how the client should use the outputs.

Document generated 12 October 2026, 22:22. For questions on methods or file locations, refer to the project folder NIGERIA_DHS_H3_PROJECT and the Stage 08 Environmental Integration directory.